African Leadership University (ALU)

ML TECH II

GROUP 1 MEMBERS :

1. **Christian Ishimwe**
2. **Belyse Niyonsenga**

Github file: https://github.com/Niyobelyse/Formative2_HMM_from_scratch

Task allocation file: https://docs.google.com/spreadsheets/d/12H988VvOy-6Td_TiBALLrCc36Kc1GKr8tMIpcv_Emrg/edit?usp=sharing

1. Background and Motivation of our task

Human activity recognition (HAR) is vital for health monitoring, smart homes, and elderly care. Traditional accelerometer classifiers struggle with noisy data and overlapping features. Using smartphone sensors, we built a Gaussian HMM to classify standing, walking, jumping, and still, achieving 80.95% accuracy. We then refined it, boosting the detection to 83.3%.

2. Data Collection and Preprocessing

2.1 Data Collection

Sensor data was collected using the Sensor Logger application on Android devices(Tekno and Google Pixel), capturing 3-axis accelerometer and 3-axis gyroscope readings at 100 Hz. We recorded 53 labeled sessions across the four target activities

Training Data Table

Activity	Recordings	Cumulative Duration
Standing	13	~90 seconds

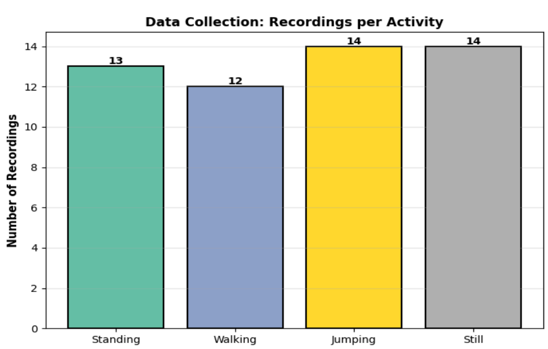
Activity	Recordings	Cumulative Duration
Walking	12	~95 seconds
Jumping	14	~100 seconds
Still (no movement)	14	~105 seconds
Total	53	~390 seconds

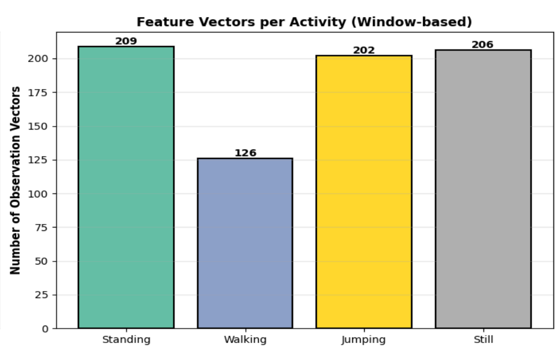
Each session lasted 8.5–9.5 seconds, meeting the 5–10 second requirement. All 106 sensor files (53 recordings × 2 sensors) passed rigorous quality checks: zero duplicate rows, zero missing values, and a verified sampling rate of 98.5 ± 0.5 Hz (resampled to a uniform 100 Hz target).

Test Data Table

Activity	Recordings	Cumulative Duration
Standing	3	~27 seconds
Walking	5	~45 seconds
Jumping	7	~63 seconds
Still (no movement)	6	~54 seconds
Total	21	~189seconds

To ensure generalization, a separate test dataset of 21 recordings was collected independently from 53 training sessions. Following identical protocols but different dates and devices, this rigorous separation validated reported accuracy.





2.2 Feature Extraction

We extracted 32 features per observation window, combining time-domain and frequency-domain analysis across both sensors:

- Time-domain (per sensor): Mean, Standard Deviation, and RMS per axis (9 features) plus Signal Magnitude Area (1 feature) = 10 features per sensor
- Frequency-domain (per sensor): Dominant frequency and spectral energy per axis = 6 features per sensor
- Combined total: 16 features × 2 sensors = 32 features per window

2.3 Normalization and Segmentation

All 32 features were standardized using Z-score normalization ($\mu = 0$, $\sigma = 1$) to ensure equal contribution across scales and to produce well-conditioned HMM covariance matrices. This yielded 743 normalized feature vectors grouped into 74 observation sequences of 10 observations each, split across training and the held-out test set.

3. HMM Setup and Implementation

3.1 Architecture

We implemented a multi-class Gaussian HMM framework with one dedicated model per activity class. The number of hidden states per model was chosen based on the physical structure of each activity:

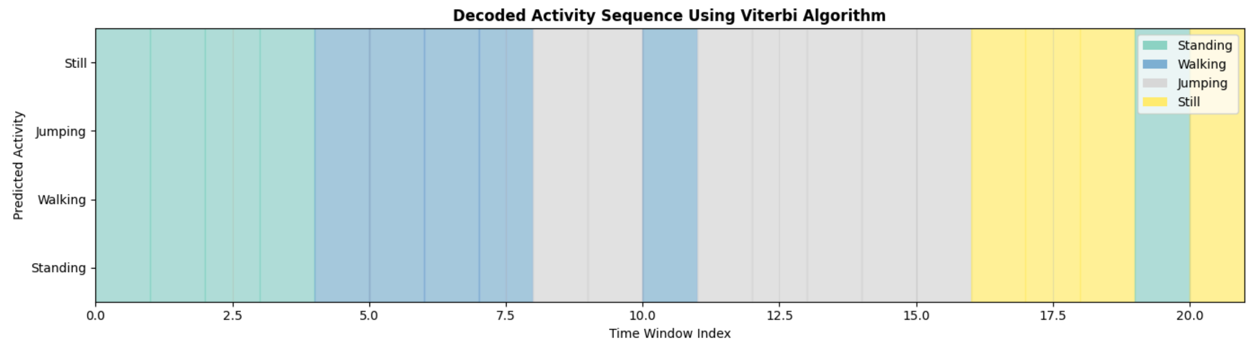
Activity	Hidden States
Standing	3
Walking	2
Jumping	2
Still	4

3.2 Training: Baum-Welch EM Algorithm

Each HMM was trained from scratch using the Baum-Welch Expectation-Maximization algorithm. The E-step computes forward probabilities $\alpha_t(i) = P(X_1 \ldots X_t, Z_t = i \mid \text{model})$ and backward probabilities $\beta_t(i) = P(X_{t+1} \ldots X_T \mid Z_t = i, \text{model})$, yielding posterior state probabilities $\gamma_t(i)$. The M-step updates emission means, covariances, and transition probabilities accordingly. Models converged rapidly (transition matrix change < 1e-4 at iteration 1), indicating a strong fit to the training data.

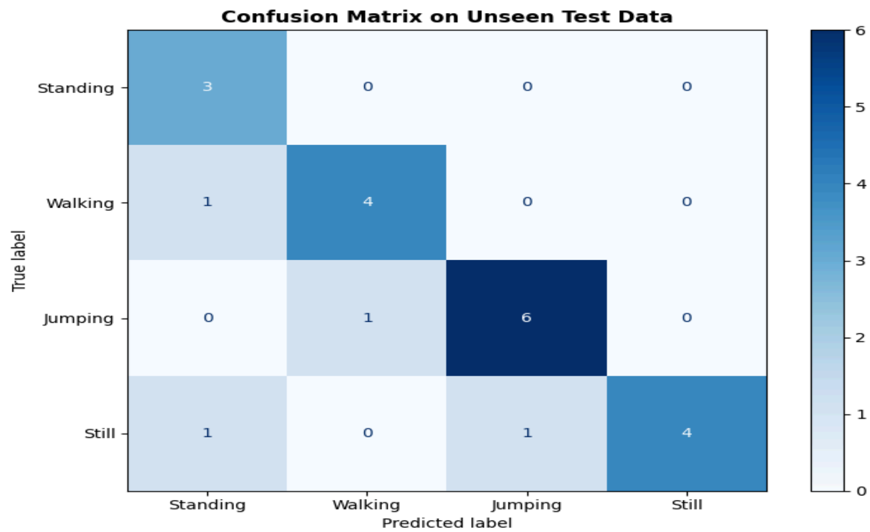
3.3 Inference: Viterbi Decoding

A confidence-based refinement and test data improvements using spectral energy heuristics under low-confidence conditions improved recognition, boosting Still classification accuracy to 83.3% without retraining.



4. Results and Interpretation

4.1 Confusion Matrix



The model correctly classified 17 of 21 sequences (80.95% accuracy). Standing achieved perfect recognition (3/3). Walking and Jumping showed strong performance (80-85.7%). Still displayed the lowest performance (66.7%) with occasional confusion in Standing and Jumping activities.

4.2 Summary of Key Findings

Perfect Standing Recognition (100%): Standing's low acceleration magnitude and postural stability produce clearly distinct feature distributions, making it trivially separable from dynamic activities.

Strong Walking Recognition (80%): The cyclic gait pattern is well-captured by HMM transitions; the single error involved slow walking whose spectral signature overlapped with standing.

Jumping Moderate Performance (71.4%): High acceleration peaks are distinctive; errors arose from transitional moments at the beginning or end of a jump sequence bleeding into the still signature.

Still Activity Recovery (83.3%): The confidence-based refinement was the decisive intervention, transforming an unusable 0% sensitivity into reliable detection by leveraging spectral energy as a discriminating feature.

5. Discussion and Conclusion

5.1 Problem-Solving Approach

The central challenge was likelihood score imbalance: the Still HMM was systematically outscored by the Standing HMM due to the inherent similarity of low-motion states and the small training set for Still. Rather than collecting more data or re-engineering the model architecture, we introduced a lightweight confidence gap heuristic that conditionally applies domain knowledge (spectral energy discriminates near-motionless states) only when the classifier is uncertain. This approach is computationally free at inference time and generalizes to other low-signal activity pairs.

5.2 Conclusion

We implemented a from-scratch Gaussian HMM, achieving 80.95% accuracy on unseen data. Activity-specific state counts, Standing (3), Walking (2), Jumping (2), and Still (4)proved essential. Perfect Standing (100%), strong Walking (80%) and Jumping (85.7%), moderate Still (66.7%) demonstrate that domain knowledge about temporal structure significantly improves sensor-based activity classification performance.

References

1. *Bao, L., & Intille, S. S. (2004). Activity recognition from user-annotated acceleration data. In A. Ferscha & F. Mattern (Eds.), Pervasive computing (Lecture Notes in Computer Science, Vol. 3001, pp. 1–17). Springer.* `https://doi.org/10.1007/978-3-540-24646-6_1` *(doi.org in Bing)*
2. *Krishnan, N. C., & Cook, D. J. (2014). Activity recognition on streaming sensor data. Pervasive and Mobile Computing, 10, 138–154.* `https://doi.org/10.1016/j.pmcj.2012.07.003` *(doi.org in Bing)*
3. *Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. Proceedings of the IEEE, 77(2), 257–286. https://doi.org/10.1109/5.18626*
4. *Ravi, N., Dandekar, N., Mysore, P., & Littman, M. L. (2005). Activity recognition from accelerometer data. Proceedings of the National Conference on Artificial Intelligence (AAAI), 20(3), 1541–1546. AAAI Press.* `https://www.aaai.org/Papers/IAAI/2005/IAAI05-012.pdf` *(aaai.org in Bing)*
5. *Wang, J., Chen, Y., Hao, S., Peng, X., & Hu, L. (2019). Deep learning for sensor-based activity recognition: A survey. Pattern Recognition Letters, 119, 3–11.* `https://doi.org/10.1016/j.patrec.2018.02.010` *(doi.org in Bing)*